

TALENT PAYS



FOR THE SEMICONDUCTOR INDUSTRY, ENGINEERS ARE THE CORE TALENT NEEDED. IN INDIA, 2.5 MILLION STUDENTS ENROL IN ENGINEERING COURSES EVERY YEAR



ENGINEERS FROM THE ELECTRONICS AND MECHANICAL STREAMS ARE THE MOST SOUGHT AFTER TALENT IN THE CHIP-MAKING INDUSTRY



AROUND 125,000 ENGINEERS ARE ALREADY WORKING IN CHIP R&D AND DESIGN IN INDIA



GLOBALLY, AROUND 2 MILLION SKILLED WORKERS WERE WORKING DIRECTLY WITH SEMICONDUCTOR DESIGN AND MANUFACTURING FIRMS IN 2021



WITH THE DEMAND FOR SKILLED WORKERS ESTIMATED TO GROW BY 1 MILLION BY 2030, THE GLOBAL INDUSTRY WOULD NEED TO ADD MORE THAN 100,000 WORKERS ANNUALLY

SOURCE: DELOITTE, INDUSTRY



testing and packaging. And even though India doesn't have foundries or ATMP facilities yet, one area where the country is contributing actively is chip design. That means working in a fabless firm. The likely jobs are:

FABLESS According to industry experts, India currently has around 125,000 engineers—with a bachelor's, master's or PhD degree—working in various aspects of chip design and development. Typically, a BTech in Electronics and Communication Engineering (ECE) or Electronics, MTech in VLSI Design, or PhD in Semiconductor Technology, Material Science or Chemical Engineering, is required to work in firms engaged in VLSI (very large-scale integration) design.

Salaries of BTech engineers working in this segment range from ₹3-18 lakh per annum. However, large multinational (MNC) chip companies, such as, Intel, Texas Instruments, NXP, etc., pay higher salaries.

ATMP Once a chip is printed, it has to be tested and packaged before it can be shipped to manufacturers. And given the low investments required in building ATMP facilities, India is expected to see a boom in their construction, both from local and MNC players.

In ATMP plants, Integrated Circuit (IC) packaging/testing engineers, with a background in electrical or electronics engineering or material science, is the core requirement. These engineers are required to understand process technologies to test and verify processes to improve yield, evaluate and verify wafer packaging materials, and do quality control of materials and components if new products are introduced.

In India, semiconductor ATMP is expected to have a much larger base than chip manufacturing. "Employers hiring for these roles look for candidates skilled in OSAT (outsourced semiconductor assembly and test), photolithography, VLSI packaging, design, and testing, along with majors in electrical and electronics engineering or material science," says Sekhar Garisa, CEO of foundit, a job-hunting platform. However, in terms of remuneration, ATMP outfits in India pay between ₹7-14 lakh per annum, whereas global firms pay between ₹20-55 lakh per annum.

FOUNDRY The chip industry is projected to invest more than \$500 billion globally to set up 84 chip-making facilities between 2021 and 2023, per the World Fab Forecast report by global industry association SEMI. These foundries would require process engineers to conduct and manage the whole process of manufacturing wafers/chips, improve process flow, evaluate and manage